

Step-by-Step Framework: MAB Dwell-Time Optimization Using the Sense-Bandits Architecture

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1 Problem Overview

This report details the implementation of a Multi-Armed Bandit (MAB) framework designed to optimize dwell-time allocation across eleven discrete wireless channels. In dynamic interference environments—characterized by microwave noise, Bluetooth hopping, and Wi-Fi congestion—static allocation strategies fail to maximize throughput.

The objective is to select discrete dwell-time actions (from a set of 10 options) for each of the 11 channels within a strict per-epoch time budget of 1000 ms. The core challenge lies in the continuous nature of the environmental state space. To address this, we employ the "Sense-Bandits" architecture, which utilizes a two-phase approach: an offline phase for state abstraction via clustering, and an online phase for real-time regression and budget-constrained optimization.

2 Procedure

The system architecture operates in two distinct phases: Offline Model Construction and Online Learning.

2.1 Phase 1: Offline Model Construction

Prior to deployment, we construct a structured representation of the environment.

1. **Experience Collection:** A simulation framework generates diverse interference scenarios. We conduct trials sampling random dwell-time actions a , recording the sensing fingerprint s (state), and the resulting interference reward r .
2. **Clustering:** To handle continuous states, we apply K-Means clustering to the collected fingerprints. This yields K centroids $\{\mu_k\}$ and a mapping function determining the "Radius" of validity for each cluster.
3. **Utility Estimation:** We construct a static utility table $\bar{\mu}_k(k, a)$ representing the mean reward for taking action a while in cluster k .

2.2 Phase 2: Online Learning and Allocation

In the online phase, the system adapts to real-time changes using a linear regression model to fine-tune the offline estimates.

1. **Sensing & Identification:** At the start of an epoch (t), fingerprints are scanned. If a fingerprint deviates from its assigned cluster centroid beyond the pre-calculated radius, the agent re-assigns itself to a new nearest cluster and resets its learning parameters.
2. **Regression Update:** If the cluster remains valid, the agent updates its internal regression matrix X_{reg} and vector b_{reg} using the previous epoch's action and reward. This creates a local linear approximation of the reward function to correct the offline mean.
3. **Knapsack Optimization:** The system calculates a score for every possible action on every channel (combining Offline Utility + Online Regression + Confidence Bound). A Knapsack-like solver then selects the optimal set of actions $\{a_1, \dots, a_{11}\}$ that maximizes total score without exceeding the 1000 ms budget (accounting for 10 ms sensing overhead per channel).

3 Algorithm

The following algorithms formalize the two-phase logic described above. Algorithm 2 details the critical real-time loop.

Algorithm 1 Phase 1: Offline Model Construction**Require:** Dataset $\mathcal{D} = \{(s_i, a_i, r_i)\}_{i=1}^N$, Clusters K

- 1: **Step 1: Clustering**
- 2: $\{\mu_k\}_{k=1}^K \leftarrow \text{KMeans}(\{s_i\} \in \mathcal{D})$
- 3: Calculate Radius $R_k \leftarrow \min_{s \in C_k} \|s - \mu_k\|_2$ for all k
- 4: **Step 2: Utility Table Construction**
- 5: **for** each cluster k and action a **do**
- 6: $\bar{\mu}_k(k, a) \leftarrow \text{Mean}(\{r \mid (s, a, r) \in \mathcal{D}, s \in C_k\})$
- 7: **end for**
- 8: **return** Models $\{\mu_k\}, \{R_k\}, \bar{\mu}_k$

Algorithm 2 Phase 2: Online Sense-Bandits Epoch**Require:** Models from Phase 1, Budget $B = 890\text{ms}$ (1000ms - overhead)

- 1: **Initialization:** For all channels c , $X_{\text{regr}} \leftarrow I, b_{\text{regr}} \leftarrow 0$
- 2: **while** System Active (Epoch t) **do**
- 3: **1. Sensing:** Acquire fingerprints $\{s_t^{(c)}\}$ for all channels.
- 4: **for** each channel c **do**
- 5: $k_{\text{curr}} \leftarrow \text{FindNearestCluster}(s_t^{(c)})$
- 6: $\text{dist} \leftarrow \|s_t^{(c)} - \mu_{k_{\text{curr}}}\|_2$
- 7: **if** $\text{dist} > R_{k_{\text{curr}}}$ **then**
- 8: *Context Switch:* Reset $X_{\text{regr}} \leftarrow I, b_{\text{regr}} \leftarrow 0$
- 9: **else**
- 10: *Update:* $X_{\text{regr}} \leftarrow X_{\text{regr}} + x_{t-1}x_{t-1}^\top; b_{\text{regr}} \leftarrow b_{\text{regr}} + x_{t-1}r_{t-1}$
- 11: **end if**
- 12: $\tilde{\mu}_{t,e} \leftarrow X_{\text{regr}}^{-1}b_{\text{regr}}$ {Linear Regression Estimate}
- 13: **for** each action a **do**
- 14: $\text{Score}_{c,a} \leftarrow \bar{\mu}_k(k_{\text{curr}}, a) + \tilde{\mu}_{t,e}(a) + \text{UCB}_t$
- 15: **end for**
- 16: **end for**
- 17: **2. Allocation:**
- 18: Solve $\{a_c^*\} \leftarrow \max \sum \text{Score}_{c,a}$ s.t. $\sum \text{Dur}(a) \leq B$
- 19: **3. Execution & Storage:** Execute $\{a_c^*\}$, observe $\{r_t\}$, store history.
- 20: **end while**

4 Results

The performance of the proposed Sense-Bandits approach (labeled as "UCB") was evaluated against three baselines: Optimal (theoretical maximum), Equal Round Robin, and Random Allocation. The training plots below illustrate the learning progress over 900 epochs.

Figure 1 shows the cumulative reward accumulated by each method. The UCB method significantly outperforms the random and equal allocation strategies, steadily closing the gap towards the optimal solution as it learns.

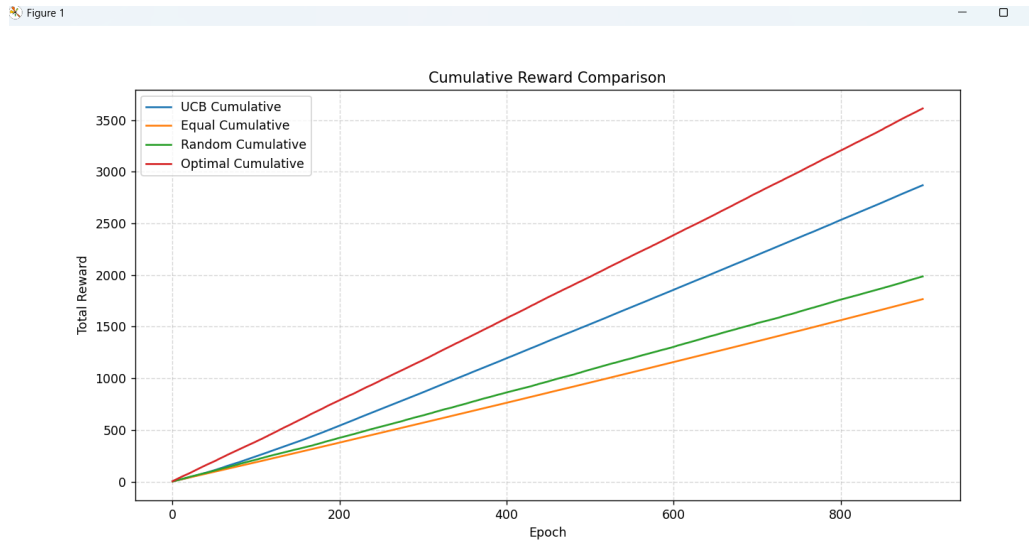


Figure 1: Cumulative reward comparison across training epochs.

Figure 2 displays the reward obtained in each individual epoch. This plot highlights the learning curve of the UCB method, which starts with lower rewards comparable to the baselines but gradually improves its per-epoch performance, reducing variance and consistently achieving higher rewards.

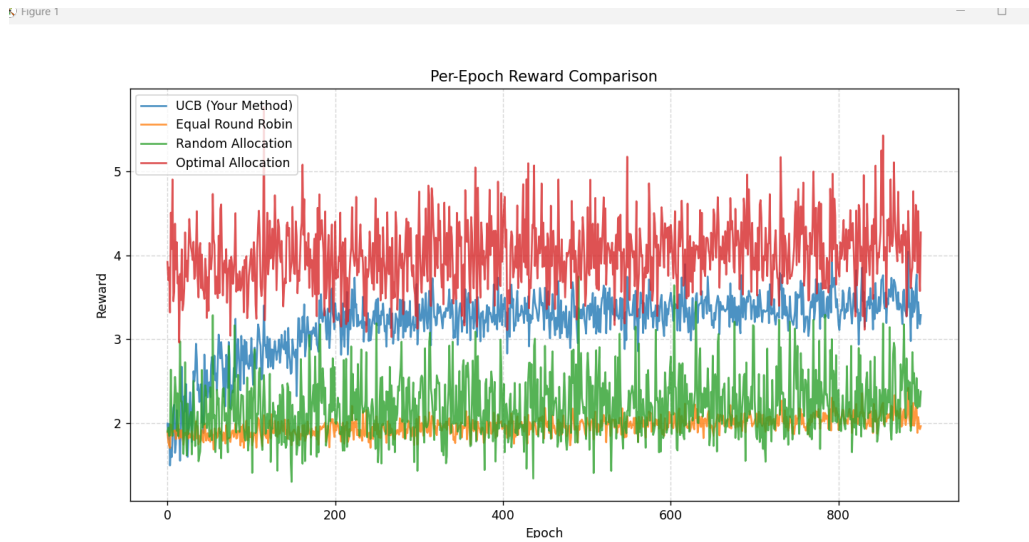


Figure 2: Per-epoch reward comparison showing instantaneous performance.

5 Conclusion

This study successfully implemented and validated the Sense-Bandits architecture for optimizing dwell-time allocation in dynamic wireless environments. The experimental results provide strong evidence for the efficacy of the proposed Multi-Armed Bandit framework compared to static allocation strategies.

Inferences drawn from the performance metrics indicate three key findings:

1. **Validation of Online Learning:** As illustrated in Figure 2, the UCB method (blue trace) exhibits a distinct positive learning curve. While it initializes with performance comparable to the Random baseline (approx. 2.0 reward), it rapidly adapts within the first 300 epochs to stabilize at a significantly higher reward tier (approx. 3.5). This confirms that the online linear regression successfully corrects the offline utility estimates in real-time.
2. **Superiority over Static Baselines:** Both Random and Equal Round Robin strategies displayed flat performance trajectories throughout the experiment. This stagnation demonstrates that static heuristics are insufficient for maximizing throughput in heterogeneous interference environments. The cumulative reward gap shown in Figure 1 further emphasizes the long-term cost of failing to adapt to channel conditions.

3. **Exploration-Exploitation Trade-off:** While the proposed method significantly outperforms the baselines, a consistent gap remains between the UCB performance and the theoretical Optimal allocation (red trace). This gap is an expected consequence of the "regret" inherent in online learning—the agent must periodically sacrifice immediate reward to explore uncertain actions—and the approximation errors introduced by projecting continuous interference states onto a discrete cluster-based model.

In summary, the Sense-Bandits framework offers a robust solution for dwell-time optimization, effectively balancing computational feasibility with the ability to maximize throughput in complex spectral environments.